Plants, Part 1: Structure & Transpiration

Test · 31 marks total

Name: ________________________________	Date: ______________	Score: _____ / 31

Section A — Plant organs & meristem tissue

A1. State one function each of the root, the stem, and the leaf.

[3 marks]

A2. What is meristem tissue?

[1 mark]

A3. Where in a plant would you find meristem tissue?

[1 mark]

A4. Explain why meristem tissue is essential for a plant to keep growing throughout its life.

[2 marks]

Section B — Xylem & phloem

B1. Name the plant tissue that transports water and dissolved mineral ions from the roots to the leaves.

[1 mark]

B2. Name the plant tissue that transports dissolved sugars around the plant.

[1 mark]

B3. State the direction(s) in which substances move in xylem, and in phloem.

[2 marks]

B4. Xylem cells are dead, with no cytoplasm, and their walls are strengthened with lignin. Explain how this helps xylem carry out its function.

[2 marks]

Section C — Transpiration

C1. Define transpiration.

[2 marks]

C2. Describe the pathway water takes from the soil to the air, starting at the root hair cell.

[4 marks]

C3. What is the name of the small pores water vapour diffuses out through, mostly on the underside of a leaf?

[1 mark]

C4. What is the function of guard cells?

[2 marks]

Section D — Factors affecting transpiration rate

D1. State four factors that affect the rate of transpiration.

[4 marks]

D2. Explain why increasing the temperature increases the rate of transpiration.

[2 marks]

D3. Explain why increasing air movement (wind) around a plant increases the rate of transpiration.

[2 marks]

D4. A student wants to measure the rate of water uptake by a cut plant shoot. Name a piece of apparatus they could use.

[1 mark]